

Haeyun Choi

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Research Interests

My research interests span various topics in computer vision, emphasizing real-world applications. I am currently focused on robust 3D scene representations and rendering techniques that remain effective under challenging conditions. The goal of my research is to move beyond idealized assumptions, embracing the complexities of real-world scenarios.

Education

MS	POSTECH , Artificial Intelligence	Mar 2023 – Feb 2025
	• Advisor: Prof. Sunghyun Cho • Thesis: "<i>Exploiting Deblurring Networks for Radiance Fields</i>"	
BS	Korea University , Biomedical Engineering	Mar 2015 – Aug 2022

Publications

Exploiting Deblurring Networks for Radiance Fields	2025
Haeyun Choi , Heemin Yang, Janghyuk Han, Sunghyun Cho IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025) Workshop on Image Processing and Image Understanding (IPIU 2025, oral)	
Optical Flow-based Shakiness Detection in Dash-cams for Hit-and-Run Accident Analysis	2024
Haeyun Choi , Sunghyun Cho Workshop on Image Processing and Image Understanding (IPIU 2024)	
AutoCycle-VC: Towards Bottleneck-Independent Zero-Shot Cross-Lingual Voice Conversion	2023
Haeyun Choi , Jio Gim, Yuho Lee, Youngin Kim, Young-Joo Suh arXiv preprint	

Awards and Honors

Outstanding Paper Award, IPIU 2025 (link)	2025
KT-POSTECH AI Expert Program Scholarship \$7,800 x 4 (link)	2023 - 2025
Excellence Prize, POSCO Youth Academy AI Competition	2022
Grand Prize, AI for Health & SDGs Competition (link)	2021
Hongsan Scholarships \$1,700	2021
Mirae Asset Exchange Student Scholarships \$3,100 (link)	2021

Work Experience

KT R&D Center , Research Scientist	Seoul, South Korea Mar 2025 – Present
• Developing an AI spatial assistant for smart glasses • Curating multimodal datasets for KT's proprietary MLLM	
POSTECH Institute of Artificial Intelligence , Research Scientist	Pohang, South Korea May 2022 – Feb 2023
• Voice conversion using mel-spectrogram representation • Video shakiness detection using optical flow statistics	

Patent

Korea Patent Application

- (P20250358) Depth-based 3D Scene Reconstruction Method From Multiple Defocus Blur Images with a Radiance-Field Model and Computing Device Thereof

Projects

Real-time Danger Alert for Convenience Stores by Motion and Voice Recognition

2022

- Developed a violence detection system for store clerks
- Applied top-down pose estimation and lightweight classification models
- Created a 20k-image dataset and enabled real-time SMS alerts with incident frames

Wear Your Mask On: Real-time Mask Detection & Warning System

2021

- Developed a mask detection system for COVID-19 public safety
- Applied MobileNet and Haar Cascade for efficient face and mask detection
- Delivered instant on-screen and audio alerts via real-time video processing

Teaching Experience

POSCO Youth AI Bigdata Academy (Video [🔗](#))

Sep 2023

- Teaching Assistant (Topic: Computer Vision)
- Led hands-on sessions and mentored AI projects

Extracurricular Activities

Team-Mate Mentorship Program (K-Digital Training [🔗](#), KCCI [🔗](#))

Aug 2024 – Present

- Selected as 1st cohort mentor [🔗](#) in a national career development program
- Peer mentorship program, supporting junior trainees in career navigation
- Engaged in job camps [🔗](#) and social contribution programs [🔗](#)

Military Service (Republic of Korea Air Force [🔗](#))

Aug 2016 – Aug 2018

- Sergeant as Operations
- Maintained 24/7 emergency standby status for mission deployment
- Managed mission planning and scheduling for search and rescue operations

Student Representative (Biomedical Engineering, Korea University)

Mar 2015 – Aug 2016

- Volunteered as student representative
- Helped organize departmental events and student activities

Skills & Languages

Skills Python, Blender, MATLAB

Languages Korean: Native, English: Fluent (TOEIC 925, TOEFL 100)